

AP CSP | Lesson 01: Bits and Binary

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. Leading zeros do not change the value of an unsigned binary integer.

Questions 1-6 | 1 mark each

Q1. Answer: D

D is correct: one byte contains eight bits, so $3 \times 8 + 1 = 25$ bits. A treats each byte as a single bit. B uses $3 + 8$ and fails to convert all three bytes or include the additional bit correctly. C counts the three bytes but omits the status bit.

Q2. Answer: B

B is correct: $00101101_2 = 32 + 8 + 4 + 1 = 45$. A, 43, would be 00101011; C, 46, would be 00101110; D, 53, would be 00110101. Each distractor places at least one 1 in a different positional weight. Leading zeros do not change the integer's value.

Q3. Answer: A

A is correct because $37 = 32 + 4 + 1$, giving 00100101. B represents $38 = 32 + 4 + 2$. C represents $41 = 32 + 8 + 1$. D represents $53 = 32 + 16 + 4 + 1$.

Q4. Answer: C

C is correct: six bits provide $2^6 = 64$ patterns, representing 0 through 63. A describes five-bit capacity. B confuses the count of patterns with the largest value and overlooks that 0 occupies a pattern. D gives the correct maximum but incorrectly uses that maximum as the count; the 0 pattern also counts.

Q5. Answer: B and D

B is correct because $2^4 = 16 < 27 \leq 32 = 2^5$. D is correct because an additional bit gives two choices for every existing pattern, increasing capacity from 32 to 64. A confuses the largest five-bit nonnegative integer, 31, with the 32 possible patterns. C confuses the two choices of a new bit with the number of additional full patterns; the increase is 32. Award one mark only for B and D together with no other choices.

Q6. Answer: B

B is correct: I exceeds a fixed integer representation's capacity, creating an overflow problem; II stores an approximation of a real value and introduces roundoff error. A reverses the concepts. C incorrectly classifies approximation as overflow; no excessive integer magnitude is stated in II. D ignores the finite representation limitations in both examples. The question does not specify or require any particular stored result after overflow.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) Eight bits; one byte. (b) $16 + 4 + 2 = 22$. (c) Yes. An agreed encoding or program context can map the pattern to a weather category; the bits alone do not determine the type or meaning of the data.

1 mark: States both eight bits and one byte.

1 mark: Shows the correct place-value sum, $16 + 4 + 2$, or an equivalent correctly weighted calculation.

1 mark: Gives the correct decimal value, 22.

1 mark: Says yes and explains that an encoding, agreed mapping, or program context determines the category meaning.

Q8. 4 marks

Answer: (a) $58 = 32 + 16 + 8 + 2$, so its binary representation is 111010. (b) 00111010. (c) 00111100 is greater: it represents 60, whereas 00111010 represents 58.

1 mark: Shows $58 = 32 + 16 + 8 + 2$, or an equivalent correct conversion method.

1 mark: Writes 111010 without leading zeros.

1 mark: Writes the correct eight-bit pattern, 00111010.

1 mark: Identifies 00111100 as greater and justifies with $60 > 58$ or a correct positional comparison.

Q9. 4 marks

Answer: (a) $2^7 = 128$. (b) $2^7 - 1 = 127$. (c) $128 - 2 = 126$. (d) Yes: 126 codes can cover 125 actions, leaving one unused action code.

1 mark: Gives $2^7 = 128$ distinct patterns.

1 mark: Gives the largest nonnegative integer as 127.

1 mark: Subtracts the two reserved patterns to obtain 126 action codes.

1 mark: States that 125 actions fit and that one available action code remains unused.

Q10. 4 marks

Answer: (a) Six bits: $2^5 = 32 < 41$ and $2^6 = 64 \geq 41$. (b) Seven bits: $2^6 = 64 < 100 \leq 128 = 2^7$. (c) $128 - 100 = 28$ unused patterns.

1 mark: Finds six bits for 41 species and shows that $2^6 = 64$ provides enough codes.

1 mark: Shows that five bits fail because $2^5 = 32 < 41$, establishing minimality.

1 mark: Finds seven bits for 100 species and supports minimality with $64 < 100 \leq 128$.

1 mark: Calculates $128 - 100 = 28$ unused patterns.

Q11. 4 marks

Answer: (a) A factor of four: $2^9 / 2^7 = 2^2 = 4$, or $512 / 128 = 4$. (b) X preserves the value. The old tag is $16 + 8 + 2 + 1 = 27$; X is also 27; Y is $64 + 32 + 8 + 4 = 108$.

1 mark: States that capacity increases by a factor of four.

1 mark: Supports the factor with $2^9 / 2^7$, $512 / 128$, or two successive doublings.

1 mark: Selects X and correctly gives both the old value and X's value as 27.

1 mark: Correctly calculates Y as 108, demonstrating that it changes the original value.

Q12. 4 marks

Answer: (a) An overflow problem: four bits represent 0 through 15, so 16 cannot be represented in this field. (b) No. Eight bits allow 0 through 255, but larger integers still exceed that fixed capacity. (c) Roundoff error: 0.142857 is only an approximation of $1/7$. (d) No. A larger finite representation can improve precision but cannot guarantee exact representation of every real number.

1 mark: Identifies overflow and explains that 16 exceeds the four-bit maximum of 15; no wraparound result is required.

1 mark: Answers no and explains that eight bits still have a finite maximum; giving 255 is valid but not required if the finite-capacity reasoning is clear.

1 mark: Identifies roundoff error and explains that the stored finite value approximates, rather than exactly equals, $1/7$.

1 mark: Answers no and explains that more bits still provide finite precision, so exact representation of every real number is not guaranteed.